

Fractal's LLM Studio will help build domain-specific AI models

Fractal Analytics has launched LLM Studio, an enterprise platform designed to build, evaluate, and operate domain-specific language models using open source foundations, marking a shift away from closed, API-led AI approaches.

The platform enables organisations to design and deploy customised LLMs using open source models, integrating selection, tuning, and benchmarking within a unified environment. It is built on NVIDIA's AI ecosystem, leveraging NVIDIA NeMo for model development and NVIDIA NIM for deployment and hosting, with plans to incorporate NVIDIA Nemotron open models.

LLM Studio introduces two core modules. AutoLLM supports open source model selection, synthetic data generation, customisation, and performance benchmarking. LLM Ops manages deployment, monitoring, and governance across the model lifecycle.

The platform focuses on enabling smaller, purpose-built models tailored to enterprise needs, offering stronger governance, predictable costs, and reliable production performance. By grounding outputs in approved enterprise data, it reduces hallucinations and improves reasoning quality. Despite using open source bases, the resulting models remain proprietary.

Built on NVIDIA reference architectures, LLM Studio standardises deployment across cloud environments, reducing the need for custom infrastructure builds.



The registry powers a rapidly expanding ecosystem of AI-native and cloud-based development environments, including Amazon's Kiro, Google's Antigravity, Cursor, IBM's Bob, VSCodium, Windsurf, and Ona (formerly Gitpod), all built on the VS Code extension API.

As usage scales, major industry players are investing in strengthening the open source infrastructure. Amazon Web Services has made a strategic investment to improve reliability, traffic management, malware detection, and platform resilience. Cursor is also supporting the registry as extension traffic grows rapidly, reflecting broader industry efforts to sustain shared open source infrastructure.

To strengthen supply-chain security, the registry has introduced a pre-publication verification framework that can detect namespace impersonation, extension name spoofing, exposed credentials, and known malicious patterns, while quarantining suspicious uploads before publication.

The platform is also transitioning to a hybrid multi-region architecture, with primary production infrastructure hosted on AWS in Europe and an independent on-premises deployment in Canada. All registry data, backups, and telemetry remain within these regions and are encrypted in transit and at rest.

NVIDIA's KVTC brings 20x memory savings to open source LLM infrastructure

NVIDIA has introduced KV Cache Transform Coding (KVTC), a new technique that reduces large language model (LLM) memory usage by up to 20x without modifying model weights or architecture, directly strengthening open source AI infrastructure.



The method delivers up to 8x faster time-to-first-token (TTFT) while maintaining less than 1% accuracy loss, even sustaining strong performance at extreme compression levels of 32x–64x. KVTC addresses a critical bottleneck in long-context AI systems, where KV cache memory can scale to multiple gigabytes, limiting GPU scalability, increasing latency, and driving infrastructure costs.

By significantly lowering memory demands, KVTC reduces GPU costs, improves prompt reuse, and enables faster multi-turn AI interactions without recomputing full conversation histories.

The approach borrows from JPEG-style media compression, combining PCA-based feature reduction, dynamic precision allocation, and DEFLATE entropy coding accelerated via NVIDIA's nvCOMP library. Compression runs between inference phases, avoiding runtime slowdowns.

"Effective KV cache management becomes critical, as idle caches must be quickly offloaded from GPU memory to accommodate other users, and quickly restored for resumed conversations," said Adrian Lancucki, senior deep learning engineer at NVIDIA. He added, "This 'media compression' approach is advantageous for enterprise deployment because it is non-intrusive: it requires no changes to model weights or code."